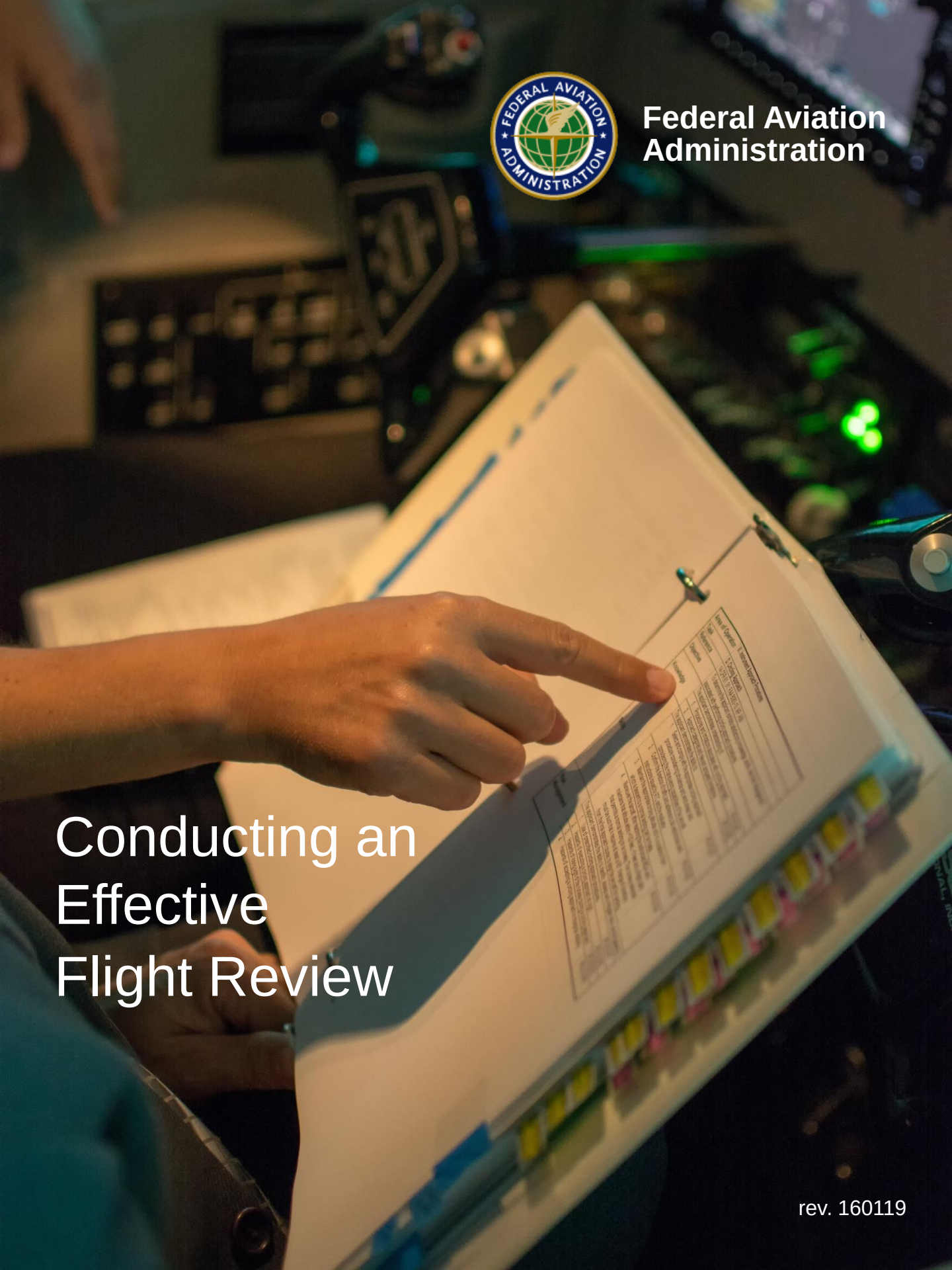


Federal Aviation Administration

Conducting an Effective Flight Review

rev. 160119

TABLE OF CONTENTS

Acknowledgements

Introduction

Step 1: Preparation

Managing Expectations
Assignments

Step 2: Ground Review

Regulatory Review
Cross-Country Flight Plan Review
Weather Decision-Making
Risk Management & Personal Minimums
General Aviation Security Issues

Step 3: Flight Activities

Physical Airplane (Basic Skills)
Mental Airplane (Systems Knowledge)
Aeronautical Decision-Making

Step 4: Post flight Debriefing

Step 5: Aeronautical Health Maintenance & Improvement

Personal Minimums Checklist
Personal Proficiency Practice Plan
Training Plan

Appendices

1 - CFI's Flight Review Checklist
2 - Pilot's Aeronautical History
3 - Regulatory Review Guide
4 - Pilot's Cross-Country Checklist
5 - Three-P Risk Management Process
6 - GA Security Checklist
7 - Personal Minimums
8 - Personal Proficiency Practice Plan
9 - Personal Training Plan
10 - Resources

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It is intended to be a living document that incorporates comments, suggestions, and ideas for best practices from GA instructors like you. Please direct comments and ideas for future iterations to: susan.parson@faa.gov.

Happy – and safe – flying! [back]

INTRODUCTION

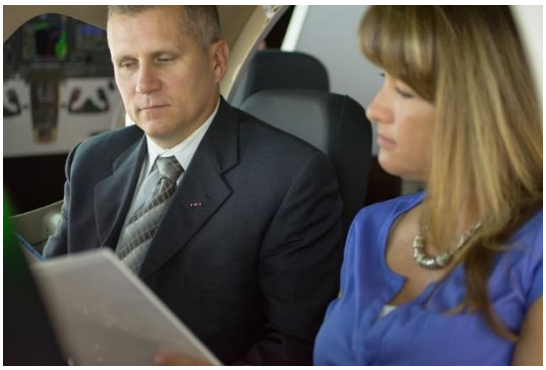


General aviation (GA) pilots enjoy a level of flexibility and freedom unrivaled by their aeronautical contemporaries. Airline, corporate, and military flight operations are all strictly regulated, and each uses a significant degree of internal oversight to ensure compliance. GA has relatively few of these regulatory encumbrances. As a result, safety depends heavily upon the development and maintenance of each individual pilot's basic skills, systems knowledge, and aeronautical decision-making skills.

The purpose of the flight review required by Title 14 of the Code of Federal Regulations (14 CFR) 61.56 is to provide for a regular evaluation of pilot skills and aeronautical knowledge. AC 61-98B states that the flight review is also intended to offer pilots the opportunity to design a personal currency and proficiency program in consultation with a certificated flight instructor (CFI). In effect, the flight review is the aeronautical equivalent of a regular medical checkup and ongoing health improvement program. Like a physical exam, a flight review may have certain "standard" features (e.g., review of specific regulations and maneuvers). However, just as the physician should tailor the exam and follow-up to the individual's characteristics and needs, the CFI should tailor both the flight review and any follow-up plan for training and proficiency to each pilot's skill, experience, aircraft, and personal flying goals.

To better accomplish these objectives, this guide, intended for use in conjunction with AC 61-98B, offers ideas for conducting an effective flight review. It also provides tools for helping that pilot develop a personalized currency, proficiency, risk management, and "aeronautical health maintenance and improvement" program. A key part of this process is the development of risk management strategies and realistic personal minimums. You can think of these minimums as individual "operations specifications" that can help guide the pilot's decisions and target areas for personal proficiency flying and future training. [back]

STEP 1: PREPARATION

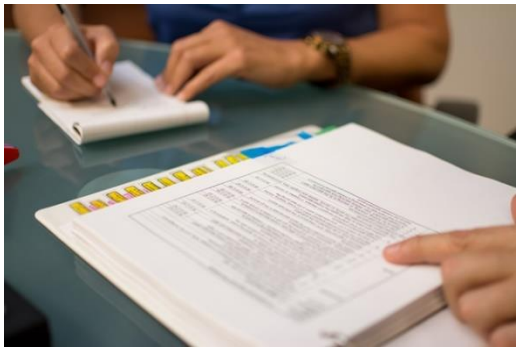


Managing Expectations: You have probably seen it, or perhaps even experienced it yourself: pilot and CFI check the clock, spend *exactly* one hour reviewing 14 CFR Part 91 operating rules, and then head out for a quick pass through the basic maneuvers generally known as "airwork." The pilot departs with a fresh flight review endorsement and, on the basis of the minimum two hours required in 14 CFR 61.56, can legally operate for the next two years. This kind of flight review may be adequate for some pilots, but for others – especially those who do not fly on a regular basis – it is not. To serve the aviation safety purpose for which it was intended, therefore, the flight review must be far more than an exercise in watching the clock and checking the box.

AC 61-98B states that the flight review is "an instructional service designed to assess a pilot's knowledge and skills." The regulations are even more specific: 14 CFR 61.56 states that the person giving the flight review has the discretion to determine the maneuvers and procedures necessary for the pilot to demonstrate "safe exercise of the privileges of the pilot certificate." It is thus a proficiency-based exercise, and it is up to you, the instructional service provider, to determine how much time and what type of instruction is required to ensure that the pilot has the necessary knowledge and skills for safe operation.

Managing pilot expectations is key to ensuring that you don't later feel pressured to conduct a "minimum time" flight review for someone whose aeronautical skills are rusty. When a pilot schedules a flight review, use the form in Appendix 2 to find out not only about total time, but also about type of flying (e.g., local leisure flying, or cross-country flying for personal transportation) and recent flight experience. You also need to know if the pilot wants to combine the flight review with a new endorsement or aircraft checkout. Offer an initial estimate of how much time to plan for ground and flight training. How much time is "enough" will vary from pilot to pilot. Someone who flies the same airplane 200 hours every year may not need as much time as someone who has logged only 20 hours since the last flight review, or a pilot seeking a new endorsement in conjunction with the flight review. For pilots who have not flown at all for several years, a useful "rule of thumb" is to plan one hour of ground training and one hour of flight training for every year the pilot has been out of the cockpit. As appropriate, you can also suggest time in an aircraft training device (ATD), or a session of night flying for pilots whose activities include flying (especially VFR) after dark.

In preparation for the flight review session, give the pilot two assignments.



Review of Part 91: The regulations (14 CFR 61.56) state that the flight review must include a review of the current general operating and flight rules set out in Part 91. The *Aeronautical Information Manual* (AIM) also contains information that pilots need to know. Have the pilot complete the Flight Review Preparation Course now available in the Aviation Learning Center at www.faasafety.gov in advance of your session and bring a copy of the completion certificate to the flight review. The online course lets the pilot review material at his or her own pace and focus attention on areas of particular interest. Alternatively, provide a copy of the list in Appendix 3 as a self-study guide.



Cross-Country Flight Plan Assignment: Many people learn to fly for personal transportation, but the cross-country flight planning skills learned for practical test purposes can become rusty if they are not used on a regular basis. Structuring the flight review as a short cross-country (i.e., 30-50 miles from the home airport) is an excellent way to refresh the pilot's flight planning skills. Ask the pilot to plan a VFR cross-country to another airport, ideally one that he or she has not previously visited. Be sure to specify that the flight plan should include consideration of runway lengths, weather, expected aircraft performance, alternatives, length of runways to be used, traffic delays, fuel requirements, terrain avoidance strategies, and NOTAM/TFR information. Proficiency in weight and balance calculations is critical as well. If the pilot regularly flies with passengers, consider asking for calculations based on maximum gross weight.

It is within your discretion to require a "manual" flight plan created with a sectional chart, plotter, and E6B. In real-world flying, however, many pilots today use tablet-based apps and online flight planning software for basic information and calculations. Appropriate use of these tools can enhance safety in several ways: they provide precise course and heading information; the convenience may encourage more consistent use of a flight plan; and automating manual calculations leaves more time to consider weather, performance, terrain, alternatives, and other aspects of the flight. Encouraging the pilot to use his or her preferred online tool will give you a more realistic picture of real-world behavior, and the computer-generated plan will give you an excellent opportunity to point out both the advantages and the potential pitfalls of this method. [back]

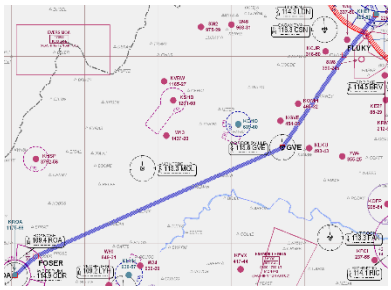
STEP 2: GROUND REVIEW

The regulations (14 CFR 61.56) specify only that the ground portion of the flight review must include "a review of the current general operating and flight rules of Part 91." This section offers guidance on conducting that review. It also provides guidance on additional topics that you should address. These include:

1. Review and discussion of the pre-assigned cross-country (XC) flight plan, with special emphasis on weather and weather decision-making; risk management and individual personal minimums; and

2. General aviation security (TFRs, aircraft security, and airport security).

Regulatory Review. Since most GA pilots do not read rules on a regular basis, this review is an important way to refresh the pilot's knowledge of information critical to aviation safety, as well as to ensure that he or she stays up to date on changes since the last flight review or formal aviation training session. If the pilot has completed the online flight review course in advance, you will want to review the results and focus primarily on those questions the pilot answered incorrectly. If the pilot has done nothing to prepare, the chart in Appendix 3 is one way to guide your discussion. You might also organize the rules as they relate to the pre-assigned cross-country flight plan that you will discuss. The important thing is to put the rules and operating procedures into a context that is relevant and meaningful to the pilot, as opposed to the sequential approach that encourages rote memorization rather than higher levels of understanding.



XC Flight Plan Review: At the most basic level, you are reviewing the pre-assigned flight plan for accuracy and completeness (i.e., are the calculations correct? Did the pilot show understanding of the 14 CFR 91.103 requirement to become familiar with "all" available information?) You may want to use the Cross-Country Checklist in Appendix 4 as a guide for checking the completeness of the pre-assigned plan.

If the pilot used automated tools to develop the flight plan, here are some questions and issues that you should teach him or her to ask about the computer-generated package:

1. How do I know that the computer-generated information is correct? (*Not all online flight planning and flight information tools are the same. Some provide real-time updates; others may be as dangerous as an out-of-date chart*.)

2. Does the computer-generated information pass the "common sense" test? (*Garbage-in, garbage-out is a fundamental principle in any kind of automation. If a pilot headed for Augusta, Georgia (KAGS) mistakenly asks for KAUG, the resulting flight plan will go to Augusta, Maine instead.*)

3. Does this plan include all the information I am required to consider? (*Some planning tools compute only course and distance, without regard to wind, terrain, performance, and other factors in a safety-focused flight plan*).

4. Does this plan keep me out of trouble? (*What if the computer-proposed course takes you through high terrain in high density altitude conditions?*)

5. What will I do if I cannot complete the flight according to this plan? (*Weather can always interfere, but pilots should also understand that flight planning software does not always generate ATC-preferred routes for IFR flying*.)

Each of these questions is directed to a critical point that you should emphasize: automated flight planning tools can be enormously helpful, but the pilot must *always* review the information with a critical eye, *frequently* supplement the computer's plan with additional information, and *never* simply assume that the computer-generated package "must be" okay because the machine is smarter.

Asking these kinds of questions is key to critical thinking, which is in turn the secret to good aeronautical decision-making (ADM) and risk management. There are many models for ADM, including charts that provide quantitative assessment and generate a numerical "score" that pilots can use in evaluating the level of risk. Although these tools can be useful, you may want to present the "3-P" method developed by the FAA Aviation Safety Program. This model encourages the pilot to **P**erceive hazards, **P**rocess risk level, and **P**erform risk management by asking a series of questions about various aspects of the flight. Appendix 5 explains this method in detail.



Since statistics show that weather is still the factor most likely to result in accidents with fatalities, the XC flight plan assignment also provides an important opportunity to discuss weather and weather decision-making. If the pilot flies VFR at night, be sure to talk about night flying considerations, especially in overcast or "no moon" conditions.

GA Security: In the post-September 11 security environment, any security incident involving general aviation pilots, aircraft, and airports can prompt calls for new

restrictions. As a flight instructor, you have a special responsibility to ensure that your clients know and follow basic security procedures. These include not only respect for temporary flight restrictions (TFRs), but also for the importance of securing your aircraft against unauthorized use. Pilots should never leave the aircraft unlocked or, worse, unattended with the keys inside.



In addition, be sure that the pilot knows about the Airport Watch Program, which was developed by the Transportation Security Administration (TSA) and the Aircraft Owners and Pilots Association (AOPA). Airport Watch relies upon the nation's pilots to observe and report suspicious activity. The Airport Watch Program is supported by a government-provided toll free hotline (1-866-GA-SECURE) and system for reporting and acting on information provided by general aviation pilots. A checklist of what to look for is in Appendix 6. For detailed information on GA security, see TSA's GA security website and AOPA's online GA security resources page.

For specific information on flying in security-restricted airspace, including the Washington DC metropolitan area Special Flight Rules Area (SFRA), direct pilots to the FAA's online SFRA training course and to the AOPA Air Safety Institute's online airspace training courses. [back]

STEP 3: FLIGHT ACTIVITIES

To operate safely in the modern flight environment, the pilot needs solid skills in three distinct, but interrelated, areas. These include:

1. "Physical Airplane" Skills (i.e., basic stick-and-rudder proficiency);

2. "Mental Airplane" Skills (i.e., knowledge and proficiency in aircraft systems);

3. Aeronautical Decision-Making (ADM) Skills (i.e., higher-order thinking skills).

Many flight reviews consist almost exclusively of airwork followed by multiple takeoffs and landings. These maneuvers can give you a very good snapshot of the pilot's "physical airplane" skills. They are also good for the pilot, who gets a safe opportunity to practice proficiency maneuvers that he or she may not have performed since the last flight review. Airwork alone, however, will tell you little about the pilot's "mental airplane" knowledge of avionics and other aircraft systems, and even less about the pilot's ability to make safe and appropriate decisions in real-world flying (ADM). Therefore, you need to structure the exercise to give you a clear picture of the pilot's skills with respect to each area.

Having the pilot fly the cross-country trip you assigned and discussed in the ground review is a good way to achieve this goal. One leg will involve flying from departure to destination, during which you ensure that the pilot encounters scenarios that let you evaluate the pilot's systems knowledge ("mental airplane") and decision-making skills, including risk management. The other leg (which can come first, depending on how you choose to organize the exercise) will focus more on airwork, which allows you to evaluate "physical airplane" skills.



Be sure to include a diversion. Remember the computer-generated flight plan discussed during the ground review portion? While you are en route to the planned destination, give the pilot a scenario that requires an immediate diversion (e.g., mechanical problem, unexpected weather). Ask the pilot to choose an alternate destination and, using all available and appropriate resources (e.g, chart, basic rules of thumb, "nearest" and "direct to" functions on the GPS) to calculate the approximate course, heading, distance, and time needed to reach the new destination. Proceed to that point and, if at all feasible, do some of the "physical airplane" pattern work at the unexpected alternate.

The diversion exercise has several benefits. First, it generates "teachable moments," which are defined as those times when the learner is most aware of the need for certain information or skills, and therefore most receptive to learning what you want to teach. Diverting to an airport surrounded by high terrain, for example, provides a "teachable moment" on the importance of obstacle awareness and terrain avoidance planning. Second, the diversion exercise quickly and efficiently reveals the pilot's level of skill in each of the three areas:

"Physical Airplane" Skills: Does the pilot maintain control of the aircraft when faced with a major distraction? For a satisfactory flight review, the pilot should be able to perform all maneuvers in accordance with the Practical Test Standards (PTS) for the pilot certificate that he or she holds.

"*Mental Airplane" Skills*: Does the pilot demonstrate knowledge and proficiency in using avionics, aircraft systems, and "bring-your-own-panel" handheld devices? Since many GA pilots use handheld GPS navigators, you will want to see whether the pilot can safely and appropriately operate the devices that will be used when you are not on board to monitor and serve as the ultimate safety net. Appropriate and proficient use of the autopilot is another "mental airplane" skill to evaluate in this exercise.

Aeronautical Decision-Making (ADM) Skills: Give the pilot multiple opportunities to make decisions. Asking questions about those decisions is an excellent way to get the information you need to evaluate ADM skills, including risk management. For example, ask the pilot to explain why the alternate airport selected for the diversion exercise is a safe and appropriate choice. What are the possible hazards, and what can the pilot do to mitigate them? Be alert to the pilot's information and automation management skills as well. For example, does the pilot perform regular "common sense cross-checks" of what the GPS and/or the autopilot are doing? [back]

STEP 4: POST FLIGHT DEBRIEFING



Most instructors have experienced the traditional "sage on the stage" model of training, in which the teacher does all the talking and hands out grades with little or no student input. There is a place for this kind of debriefing; however, a collaborative critique is one of the most effective ways to determine that the pilot has not only the physical and mental airplane skills, but also the self-awareness and judgment needed for sound aeronautical decision-making. Here is one way to structure a collaborative post flight critique:

Replay: Rather than starting the post flight briefing with a laundry list of areas for improvement, ask the pilot to verbally *replay* the flight for you. Listen for areas where your perceptions are different, and explore why they don't match. This approach gives the pilot a chance to validate his or her own perceptions, and it gives you critical insight into his or her judgment abilities.

Reconstruct: The reconstruct stage encourages the pilot to learn by identifying the "would'a could'a should'a" elements of the flight – that is, the key things that he or she *would have*, *could have*, or *should have* done differently.

Reflect: Insights come from investing perceptions and experiences with meaning, which in turn requires reflection on these events. For example:

1. What was the most important thing you learned today?
2. What part of the session was easiest for you? What part was hardest?
3. Did anything make you uncomfortable? If so, when did it occur?
4. How would you assess your performance and your decisions?
5. Did you perform in accordance with the Practical Test Standards?

Redirect: The final step is to help the pilot relate lessons learned in this flight to other experiences, and consider how they might help in future flights. Questions:

1. How does this experience relate to previous flights?
2. What might you do to mitigate a similar risk in a future flight?
3. Which aspects of this experience might apply to future flights, and how?
4. What personal minimums should you establish, and what additional proficiency flying and training might be useful? [back]

STEP 5: "AERONAUTICAL HEALTH" MAINTENANCE & IMPROVEMENT

If the pilot did not perform well enough for you to endorse him or her for satisfactory completion of the flight review, use the PTS as the objective standard to discuss areas needing improvement, as well as areas where the pilot performed well. Offer a practical course of action – ground training, flight training, or both – to help him or her get back up to standards. If possible, offer to schedule the next session before the pilot leaves the airport.

If the pilot's performance on both ground and flight portions was satisfactory, you can complete the flight review simply by endorsing the pilot's logbook. However, offer the pilot an opportunity to develop a personalized aeronautical health maintenance and improvement plan. Such a plan should include consideration of the following elements:

Personal Minimums Checklist: One of the most important concepts to convey in the flight review is that safe pilots understand the difference between what is "legal" in terms of the regulations, and what is "smart" in terms of pilot experience and proficiency. For this reason, assistance in completing a Personal Minimums Checklist tailored to the pilot's individual circumstances is perhaps the single most important "takeaway" item you can offer. Use the Personal Minimums Development Worksheets in Appendix 7 to help your client work through some of the questions that should be considered in establishing "hard" personal minimums, as well as in preflight and in-flight decision-making.

Personal Proficiency Practice Plan: Flying just for fun is one of the most wonderful benefits of being a pilot, but many pilots would appreciate your help in developing a plan for maintaining and improving basic aeronautical skills. You might use the suggested flight profile in Appendix 8 as a guide for developing a regular practice plan.

Training Plan: Discuss and schedule any additional training the pilot may need to achieve individual flying goals. For example, the pilot's goal might be to develop the competence and confidence needed to fly at night, or to lower personal minimums in one or more areas. Another goal might be completion of another phase in the FAA's Pilot Proficiency ("Wings") Program, or obtaining a complex, high performance, or tailwheel endorsement. Use the form in Appendix 9 to document the pilot's aeronautical goals and develop a specific training plan to help him or her achieve them.

The flight review is vital link in the general aviation safety chain. As a person authorized to conduct this review, you play a critical role in ensuring that it is a meaningful and effective tool for maintaining and enhancing GA safety. [back]

Appendix 1 - CFI's Flight Review Checklist

Step 1: Pre-Flight Review Actions

__Scheduling
__Pilot's Aeronautical History
__Part 91 Review Assignment
__Cross-Country Flight Plan Assignment

Step 2: Ground Discussion

__Regulatory Review
__Cross-Country Flight Plan Review
__Risk Management & Personal Minimums

Step 3: Conducting the Flight

__Physical Airplane (basic skills)
__Mental Airplane (systems knowledge)
__Aeronautical Decision-Making

Step 4: Postflight Discussion

__Replay, Reflect, Reconstruct, Redirect
__Questions

Step 5: Aeronautical Health Maintenance & Improvement Plan

__Personal Minimums Checklist
__Personal Proficiency Practice Plan
__Training Plan (if desired)
__Resources List

[back]

Appendix 2 - Pilot's Aeronautical History for Flight Review

Pilot's Name:_____ CFI:_____
Address:_____
Phone(s):_____ e-mail:_____

Type of Pilot Certificate(s):
Private_____ Commercial_____ ATP_____Flight Instructor_____

Rating(s):
Instrument_____ Multiengine _____

Experience (Pilot):
Total time_____ Last 6 months_____ Avg hours/month_____

Time logged since last flight review_____ Since last IPC_____

Experience (Aircraft):
Aircraft type(s) you fly_____
_____

Aircraft used most often_____

For this aircraft:
Total time_____ Last 6 months_____ Avg hours/month_____

Experience (Flight environment):

Since your last flight review, approximately how many hours have you logged in:

Day VFR_____ Day IFR_____ IMC_____
Night VFR_____ Night IFR_____

Mountainous terrain_____ Overwater flying_____

Airport with control tower_____ Airport w/o control tower_____

Type of Flying (External factors):

What percentage of your flying is for:
Pleasure_____ Business_____Local_____ XC_____

Personal Skills Assessment:
What are your strengths as a pilot?_____
What do you most want to practice/improve?_____
What are your aviation goals?_____ [back]

Appendix 3 - Regulatory Review Guide

Pilot

Experience:
Recent flight experience (61.57)

Responsibility:
Authority (91.3)
ATC Instructions (91.123)
Preflight action (91.103)
Safety belts (91.107)
Flight crew at station (91.105)

Cautions:
Careless or reckless operation (91.13)
Dropping objects (91.15)
Alcohol or drugs (91.17
Supplemental oxygen (91.211)
Fitness for flight (AIM Chapter 8, Section 1)

Aircraft

Airworthiness:
Basic (91.7)
Flight manual, markings, placards (91.9)
Certifications required (91.203)
Instrument & equipment requirements (91.205)
-ELT (91.207)
-Position lights (91.209)
-Transponder requirements (91.215)
-Inoperative instruments and equipment (91.213)

Maintenance:
Responsibility (91.403)
Maintenance required (91.405)
Maintenance records (91.417)
Operation after maintenance (91.407)
Inspections:
Annual, Airworthiness Directives, 100-Hour (91.409)
Altimeter & Pitot Static System (91.411)
VOR check (91.171)
Transponder (91.413) & ELT (91.207)

enVironment

Airports
Markings (AIM Chapter 2, Section 3)
Operations (AIM 4-3; 91.126, 91.125)
Traffic Patterns (91.126

Airspace
Altimeter Settings (91.121; AIM 7-2)
Minimum Safe Altitudes (91.119, 91.177)
Cruising Altitudes (91.159, 91.179; AIM 3-1-5)
Speed Limits (91.117)
Right of Way (91.113)
Formation (91.111)
Types of Airspace (AIM 3)
-Controlled Airspace (AIM 3-2; 91.135, 91.131, 91.130, 91.129)
-Class G Airspace (AIM 3-3)
-Special Use (AIM 3-4; 91.133, 91.137, 91.141. 91.143, 91.145)
Emergency Air Traffic Rules (91.139; AIM 5-6)

Air Traffic Control & Procedures
Services (4-1)
Radio Communications (4-2 & Pilot/Controller Glossary)
Clearances (4-4)
Procedures (AIM 5)

Weather
Meteorology (AIM 7-1)
Wake Turbulence (AIM 7-3)

External Pressures

Personal Minimums Checklist [back]

Appendix 4 - Pilot's Cross-Country Checklist

PILOT

__Review Personal Minimums Checklist
- __Recency (time/practice in last 30 days)
- __Currency (takeoffs & landings, IFR currency if applicable)
- __Terrain & airspace (familiarity?)
- __Health & well-being

AIRCRAFT

__Overall mechanical condition
__Avionics & systems
__Performance calculations
__Fuel requirements
__Other equipment

ENVIRONMENT

__Weather Reports & Forecasts
- __Departure/En route/Destination
- __Severe weather forecasts?
- __Weather stability?
- __Alternate required?

__Night
- __Flashlights available
- __Terrain avoidance plan

__Airspace
- __TFRs or other restrictions
- __COM/NAV equipment requirements
- __Cruising altitude(s)

__Terrain
- __VFR & IFR charts with MSA / MEA altitudes
- __AOPA/ASI Terrain Avoidance Planning

__Airports
- __COM/NAV requirements & frequencies
- __Runway lengths
- __Services available

EXTERNAL PRESSURES

__Family expectations?
__Passenger needs / expectations?
__Weather worries?
__Prepared for diversion (money, accommodations)?
__Time pressures (e.g., "must be at work" issues)? [back]

Appendix 5 - Three-P Risk Management Process

Good aeronautical decision-making includes risk management, a process that systematically identifies hazards, assesses the degree of risk, and determines the best course of action. There are many models for risk management, including charts that generate a numerical "score." Although these tools can be useful, numbers-based tools suggest a level of precision that may be misleading.

An alternative method is the Perceive – Process – Perform risk management and aeronautical decision-making model developed by the FAA Aviation Safety Program. There are three basic steps in this model:



PERCEIVE hazards
PROCESS to evaluate level of risk
PERFORM risk management

PERCEIVE: The goal is to identify hazards, which are events, objects, or circumstances that could contribute to an undesired event. You need to consider hazards associated with:

Pilot
Aircraft
en**V**ironment
External Pressures.

PROCESS: Ask questions to determine what can hurt you. In short, why do you have to **CARE** about these hazards?

What are the **C**onsequences?
What are the **A**lternatives available to me?
What is the **R**eality of the situation facing me?
What kind of **E**xternal pressures may affect my thinking?

PERFORM: Change the situation in your favor. Your objective is to make sure the hazard does not hurt **ME** or my loved ones, so work to either

Mitigate the risk involved, or
Eliminate the risk involved. [back]

Appendix 6 - General Aviation Security

The Transportation Security Administration (TSA) has partnered with the Aircraft Owners and Pilots Association (AOPA) to develop a nationwide Airport Watch Program that uses the more than 650,000 pilots as eyes and ears for observing and reporting suspicious activity. This partnership helps general aviation keep our airports secure without needless and expensive security requirements. AOPA Airport Watch is supported by a centralized government provided toll free hotline (1-866-GA-SECURE) and system for reporting and acting on information provided by general aviation pilots. The Airport Watch Program includes warning signs for airports, informational literature, and training videotape to educate pilots and airport employees as to how security of their airports and aircraft can be enhanced.

Here's what to look for:

1. Pilots who appear under the control of someone else.
2. Anyone trying to access an aircraft through force — without keys, using a crowbar or screwdriver.
3. Anyone who seems unfamiliar with aviation procedures trying to check out an airplane.
4. Anyone who misuses aviation lingo — or seems too eager to use all the lingo
5. People or groups who seem determined to keep to themselves.
6. Any members of your airport neighborhood who work to avoid contact with you or other airport tenants.
7. Anyone who appears to be just loitering, with no specific reason for being there.
8. Any out-of-the-ordinary videotaping of aircraft or hangars.
9. Aircraft with unusual or obviously unauthorized modifications.
10. Dangerous cargo or loads — explosives, chemicals, openly displayed weapons — being loaded into an airplane.
11. Anything that strikes you as wrong — listen to your gut instinct, and then follow through.
12. Pay special attention to height, weight, and the individual's clothing or other identifiable traits.

Use common sense. Not all these items indicate terrorist activity.

When in doubt, check it out! Check with airport staff or call the National Response Center 1-866-GA-SECURE! [back]

Appendix 7 - Personal Minimums – Decision Making in Advance

One of the most useful things a pilot can do in aviation safety risk management is to develop and write down personal minimums. In formal terms, personal minimums are an individual pilot's set of procedures, rules, criteria, and guidelines for deciding whether, and under what conditions, to operate (or continue operating). While accurate, the formal definition does not really convey one of the core concepts: personal minimums as a "safety buffer" between the demands of the situation and the extent of both pilot skills and airplane performance.

Think of personal minimums as the human factors equivalent of reserve fuel. When the pilot plans a flight, the regulations require calculating fuel use in a way that leaves a specified amount of fuel in the tanks upon landing. Reserve fuel is intended to provide a safety buffer between fuel *required* for normal flight and fuel *available* to avoid total quiet in the engine compartment.

In the same way, a pilot should establish written personal minimums to provide a solid safety buffer between the skills and aircraft performance *required* for a specific flight, and the skills and aircraft performance *available*.

Does your pilot have written personal minimums? If not, one of the most helpful things you can do is to encourage him or her to invest the time in developing them. For one approach to this process, you can point your pilot to the "Getting the Maximum from Personal Minimums" article from the May/June 2006 of *FAA Aviation News* (http://www.faa.gov/news/safety_briefing/2006/media/mayjun2006.pdf). The article provides a step-by-step approach and worksheets the pilot can use for this process.

If the pilot does have written personal minimums, you might ask whether the document is up-to-date. Personal minimums are very dynamic, because proficiency levels change (for better and for worse) in accordance with practice.

Once personal minimums have been established and updated, a right seat passenger can contribute to good risk management by asking the pilot to demonstrate that the proposed flight is consistent with those pre-established decisions. In addition to increasing the passenger's level of comfort and confidence, this approach makes it easier for the pilot to make "disappointing" decisions when circumstances so require. [back]

Appendix 8 - Personal Proficiency Practice Plan

Pilot's Name:_____
CFI:_____
Date:_____
Review Date:_____

VFR Flight Profile – Every 4-6 Weeks:

Preflight (include 3-P Risk Management Process)

Normal taxi, takeoff, departure to practice area.

CHAPS (before each maneuver):
- **C**lear the area
- **H**eading established & noted
- **A**ltitude established (at least 3,000 AGL)
- **P**osition near a suitable emergency landing area
- **S**et power and aircraft configuration

Steep turns (both directions), maintaining altitude w/i 100' and airspeed w/i 10 knots.

Power-off stalls (approach to landing) & recovery.

Power-on stalls (takeoff/departure) & recovery.

Ground reference maneuvers.

Pattern practice:
- Normal landing (full flaps)
- Short-field takeoff and landing over a 50' obstacle
- Soft-field takeoff and landing

Secure the aircraft.

Review your performance.

Schedule next proficiency flight.
[back]

Appendix 9 - Personal Aeronautical Goals

Pilot's Name:_____ CFI:_____
Date:_____ Review Date:_____

Training Goals

_____ Certificate Level (Private, Commercial, ATP)
_____ Ratings (Instrument, AMEL, ASES, AMES, etc)
_____ Endorsements (high performance, complex, tailwheel, high altitude)
_____ Phase in Pilot Proficiency (WINGS) Program
_____ Instructor Qualifications (CFI, CFI-I, MEI, AGI, IGI)
Other: _____
_____

Proficiency Goals

_____ Lower personal minimums to
- _____ Ceiling
- _____ Visibility
- _____ Winds
- _____ Precision Approach Minimums
- _____ Non-Precision Approach Minimums

_____ Fly at least:
- _____ Times per month
- _____ Hours per month
- _____ Hours per year
- _____ XC flights per year
- _____ Night hours per month

_____ Make a XC trip to:
_____

Other: _____
_____

Aeronautical Training Plan

_____
_____
_____
_____
_____[back]

Appendix 10 - Resources

Airman Certification Standards
http://www.faa.gov/training_testing/testing/

Airman Testing Standards & Training
http://www.faa.gov/training_testing/testing/

Currency and Additional Qualification Requirements for Certificated Pilots (AC 61-98B)
www.faa.gov/documentLibrary/media/Advisory_Circular/AC%2061-98B.pdf

Instrument Proficiency Check Guidance
http://www.faa.gov/pilots/training/media/IPC_Guidance.pdf

Best Practices for Mentoring in Flight Instruction
http://www.faa.gov/training_testing/training/media/mentoring_best_practices.pdf

FAA Safety Briefing
https://www.faa.gov/news/safety_briefing/

- Airspace and ATC – Jan/Feb 2015
- Weather Forces, Sources, and Resources – Mar/Apr 2015
- New Technology in Aviation – Jan/Feb 2014
- Getting Back in the Game – MarApr 2014
- Flying Companion's Guide to GA – JulAug 2014

FAA Safety Team (FAASTeam)
www.faasafety.gov

Security for GA
www.tsa.gov

Security for GA (AOPA Airport Watch)
www.aopa.org

[back]